

LAB EXPERIMENT RECORD

Set 1: Monthly Sales Data

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To visualize monthly sales data using a line chart, bar chart, and scatter plot, and to build an interactive dashboard using R and Shiny.

Dataset: Monthly Sales

Month	Sales (\$)
January	15000
February	18000
March	22000
April	20000
May	23000

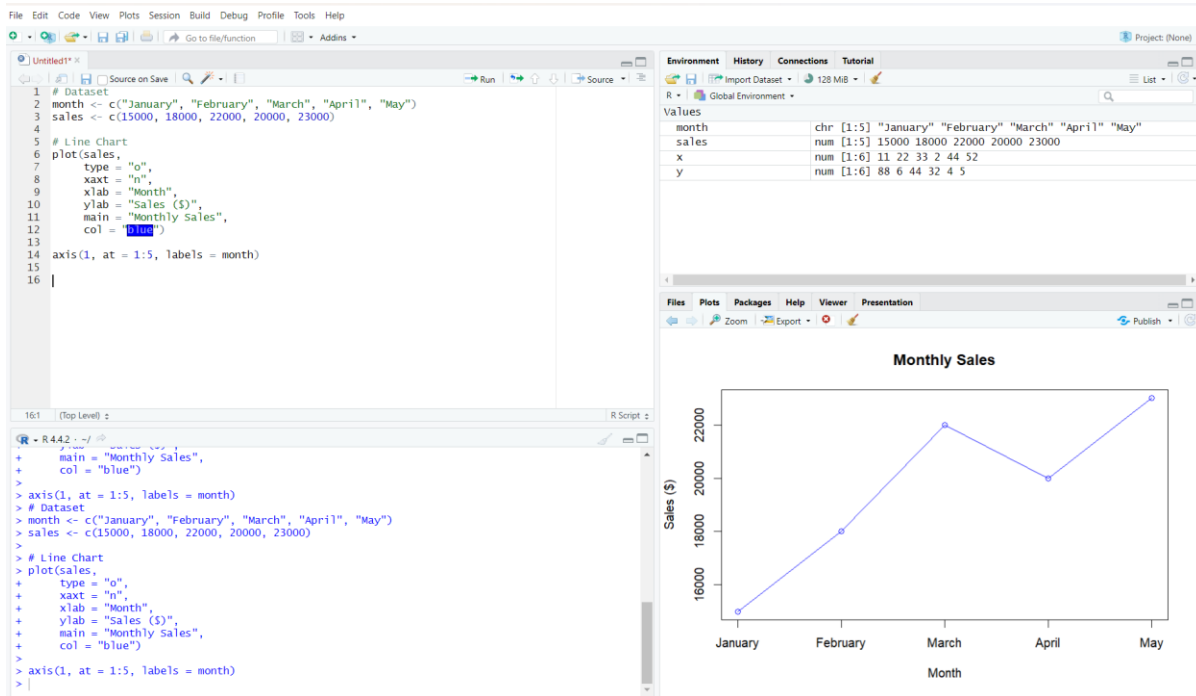
PROGRAM 1: Line Chart of Monthly Sales

Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

# Line Chart
plot(sales,
     type = "o",
     xaxt = "n",
     xlab = "Month",
     ylab = "Sales ($)",
     main = "Monthly Sales",
     col = "blue")

axis(1, at = 1:5, labels = month)
```



Output 1: Monthly Sales Line Chart

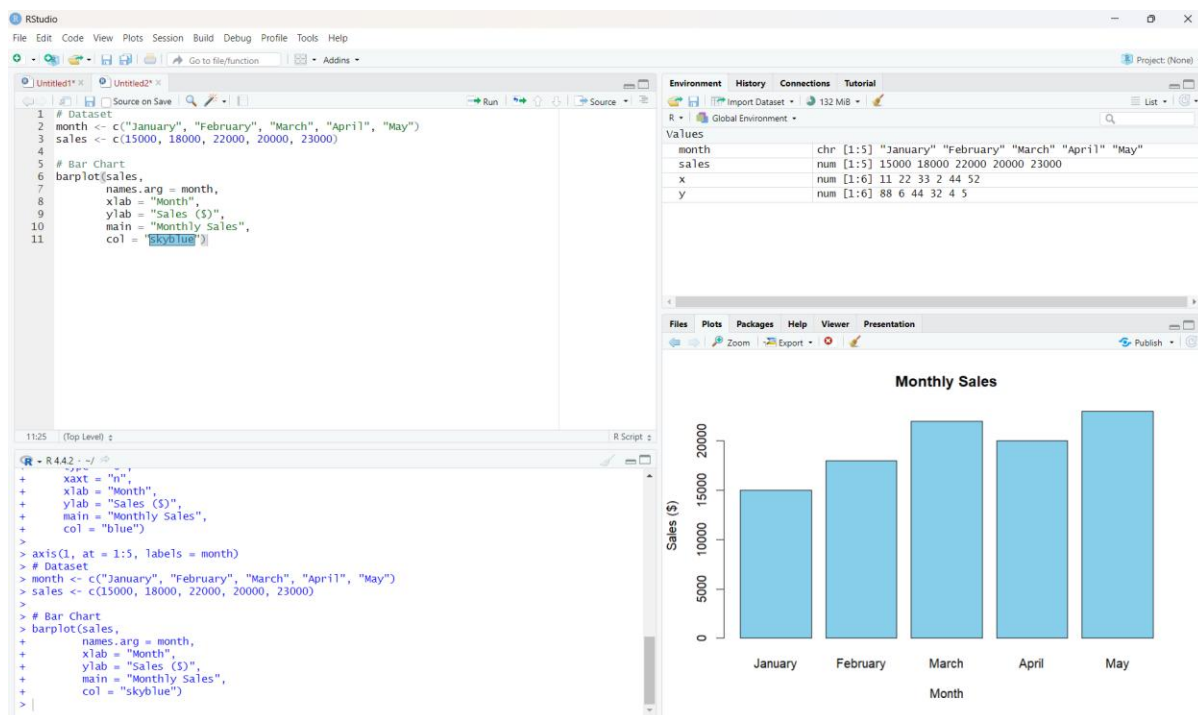
Observation: Sales increase from January to March, decrease slightly in April, and reach the highest value in May.

PROGRAM 2: Bar Chart of Sales

Generate a bar chart to display the sales values for the available months and label the chart elements.

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

# Bar Chart
barplot(sales,
        names.arg = month,
        xlab = "Month",
        ylab = "Sales ($)",
        main = "Monthly Sales",
        col = "skyblue")
```



Output 2: Monthly Sales Bar Chart

Observation: The bar chart makes it easy to compare the sales of each month. May has the highest sales and January has the lowest.

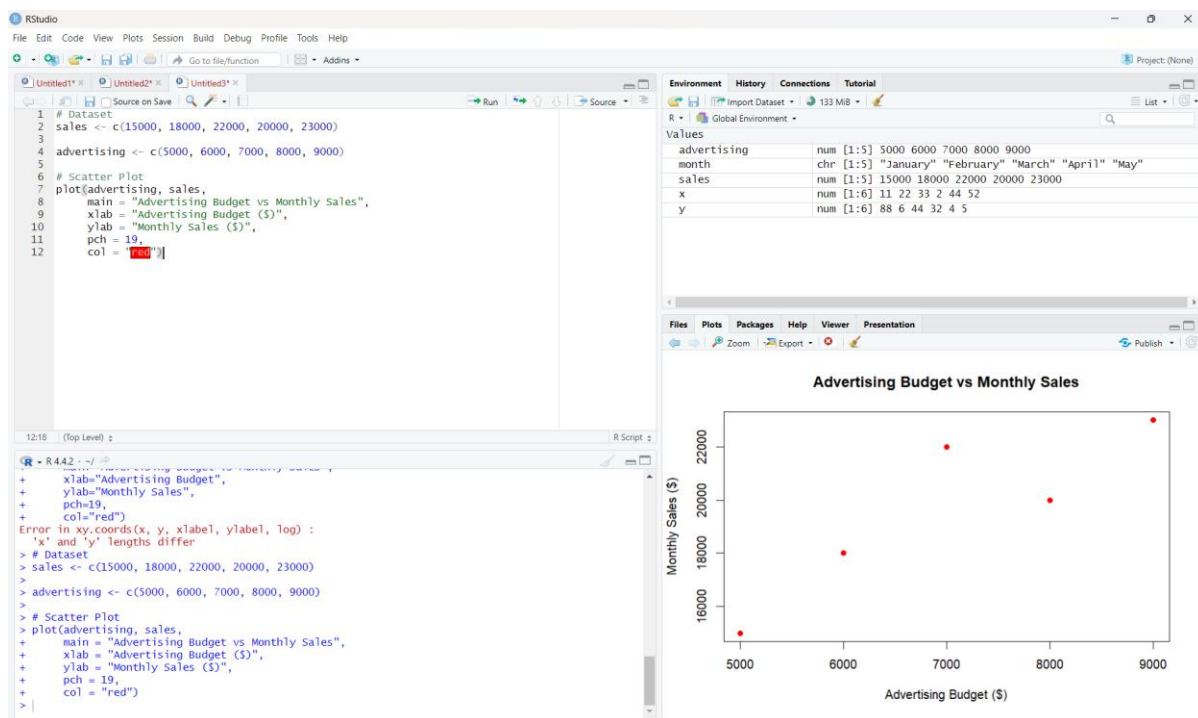
PROGRAM 3: Scatter Plot of Advertising Budget vs Monthly Sales

Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

```
# Dataset
sales <- c(15000, 18000, 22000, 20000, 23000)

advertising <- c(5000, 6000, 7000, 8000, 9000)

# Scatter Plot
plot(advertising, sales,
     main = "Advertising Budget vs Monthly Sales",
     xlab = "Advertising Budget ($)",
     ylab = "Monthly Sales ($)",
     pch = 19,
     col = "red")
```



Output 3: Advertising Budget vs Monthly Sales Scatter Plot

Insight: The plotted points show a generally positive relationship: as the advertising budget increases from \$5,000 to \$9,000, monthly sales also generally increase. This suggests that higher advertising spending is associated with higher sales for the values plotted.

Note: The visible assignment table contains only Month and Sales. The advertising values used in the submitted R program are the five values shown in the user's executed program: 5000, 6000, 7000, 8000, and 9000.

PROGRAM 4: Interactive Sales Dashboard

Build an interactive dashboard combining the line chart and bar chart using the Shiny package.

```
library(shiny)

month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

ui <- fluidPage(

  titlePanel("Monthly Sales Dashboard"),

  sidebarLayout(

    sidebarPanel(
      selectInput("chart",
                  "Select Chart:",
                  choices = c("Line Chart", "Bar Chart"))
    ),

    mainPanel(
      plotOutput("salesPlot")
    )
  )
)

server <- function(input, output) {

  output$salesPlot <- renderPlot({

    if (input$chart == "Line Chart") {

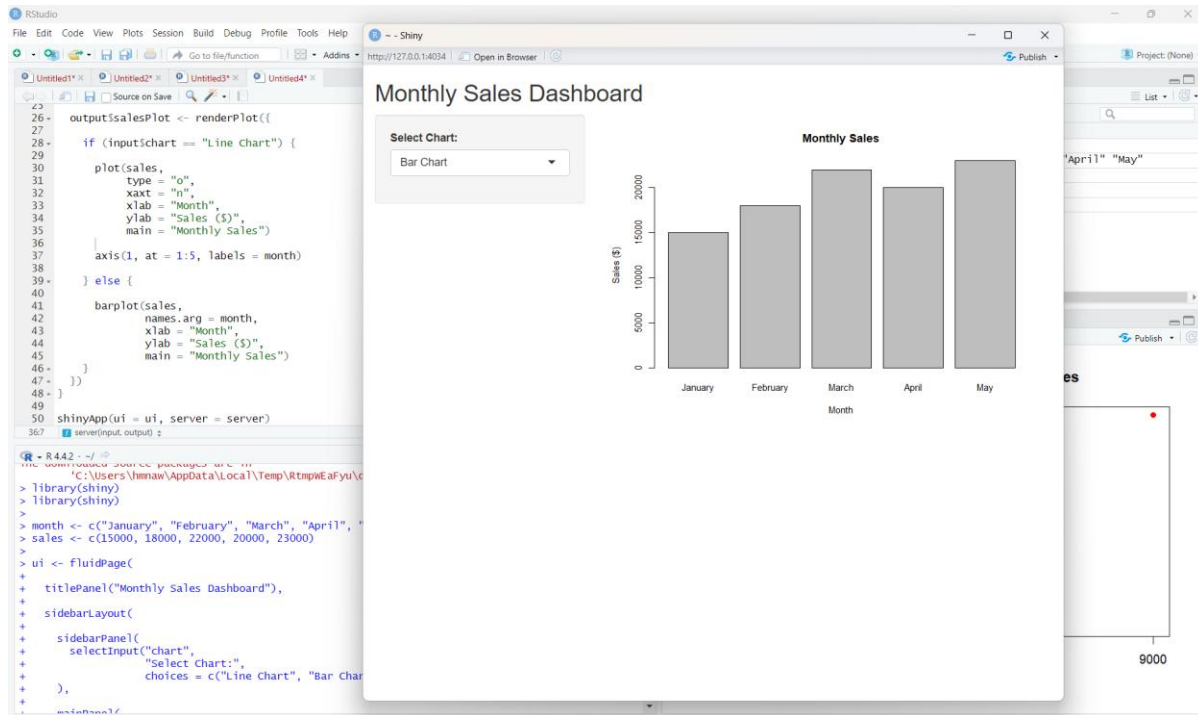
      plot(sales,
           type = "o",
           xaxt = "n",
           xlab = "Month",
           ylab = "Sales ($)",
           main = "Monthly Sales")

      axis(1, at = 1:5, labels = month)

    } else {

      barplot(sales,
              names.arg = month,
              xlab = "Month",
              ylab = "Sales ($)",
              main = "Monthly Sales")
    }
  })
}

shinyApp(ui = ui, server = server)
```



Output 4: Interactive Monthly Sales Dashboard

Observation: The dashboard provides a chart-selection control. Selecting Line Chart displays the monthly trend, while selecting Bar Chart displays the monthly sales comparison.

Procedure

1. Create the required month and sales vectors in R.
2. Use `plot()` to create the line chart with appropriate axis labels and title.
3. Use `barplot()` to compare the sales values for the available months.
4. Create advertising-budget and sales vectors of equal length and use `plot()` for the scatter plot.
5. Install and load the Shiny package, then create the UI and server components for the interactive dashboard.
6. Run the Shiny application and switch between the Line Chart and Bar Chart using the selection box.

Result

The monthly sales data was successfully visualized using line, bar, and scatter charts. An interactive Shiny dashboard was also created to switch between the line chart and bar chart.

Key R Functions Used

Function	Purpose
<code>plot()</code>	Creates line and scatter plots.
<code>axis()</code>	Adds custom axis labels.
<code>barplot()</code>	Creates bar charts.
<code>fluidPage()</code>	Creates the Shiny page layout.
<code>selectInput()</code>	Provides the chart selection control.
<code>renderPlot()</code>	Generates plots dynamically in Shiny.
<code>shinyApp()</code>	Runs the interactive Shiny application.

SET 2: CUSTOMER FEEDBACK ANALYSIS

Dataset: Customer Satisfaction

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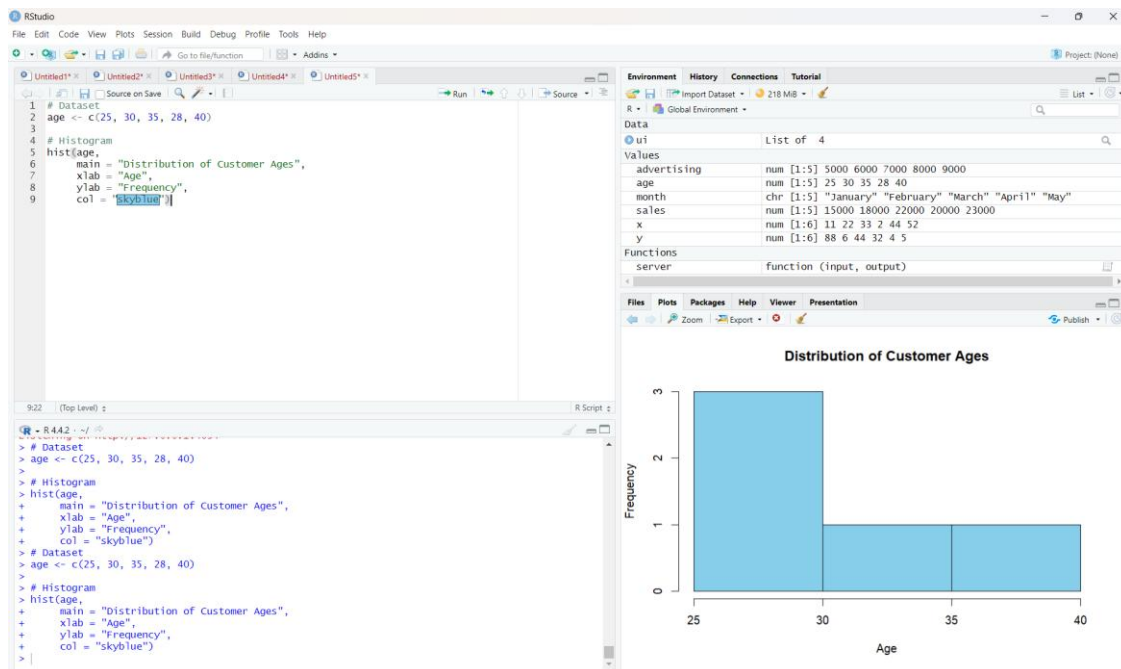
Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

PROGRAM 1: Histogram of Customer Ages

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
# Dataset
age <- c(25, 30, 35, 28, 40)

# Histogram
hist(age,
      main = "Distribution of Customer Ages",
      xlab = "Age",
      ylab = "Frequency",
      col = "skyblue")
```



Output 1: Distribution of Customer Ages

Observation: The histogram represents the distribution of the five customer ages in the given dataset.

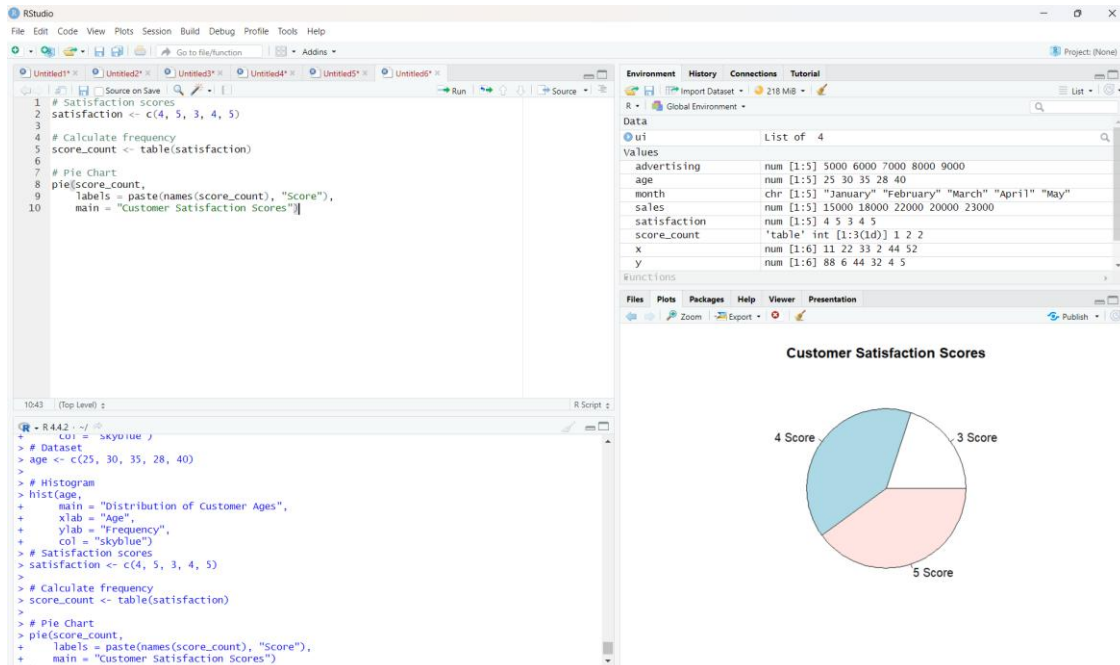
PROGRAM 2: Pie Chart of Satisfaction Scores

Generate a pie chart to display the overall distribution of customer satisfaction scores and include labels.

```
# Satisfaction scores
satisfaction <- c(4, 5, 3, 4, 5)

# Calculate frequency
score_count <- table(satisfaction)

# Pie Chart
pie(score_count,
    labels = paste(names(score_count), "Score"),
    main = "Customer Satisfaction Scores")
```



Output 2: Customer Satisfaction Scores Pie Chart

Observation: Scores 4 and 5 each occur twice, while score 3 occurs once.

PROGRAM 3: Stacked Bar Chart by Age Group

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
# Dataset
age <- c(25, 30, 35, 28, 40)
satisfaction <- c(4, 5, 3, 4, 5)

# Create age groups
age_group <- cut(age,
    breaks = c(20, 29, 39, 49),
    labels = c("20-29", "30-39", "40-49"))

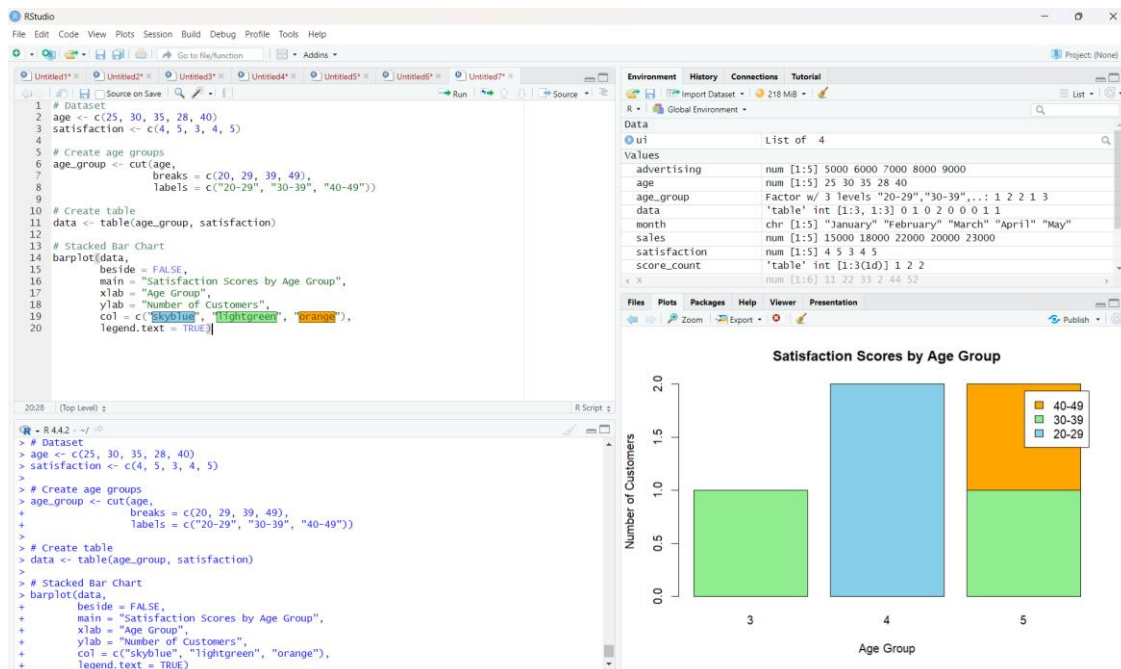
# Create table
data <- table(age_group, satisfaction)

# Stacked Bar Chart
barplot(data,
    beside = FALSE,
```

```

main = "Satisfaction Scores by Age Group",
xlab = "Age Group",
ylab = "Number of Customers",
col = c("skyblue", "lightgreen", "orange"),
legend.text = TRUE)

```



Output 3: Satisfaction Scores by Age Group

Observation: The code creates three age groups and stacks the satisfaction-score frequencies for each group.

Note: The supplied RStudio screenshot is inserted exactly as captured, including its displayed x-axis labels.

PROGRAM 4: Word Cloud of Customer Feedback

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```

library(wordcloud)
library(RColorBrewer)

feedback <- c(
  "good service",
  "excellent service",
  "good experience",
  "friendly staff",
  "excellent service"
)

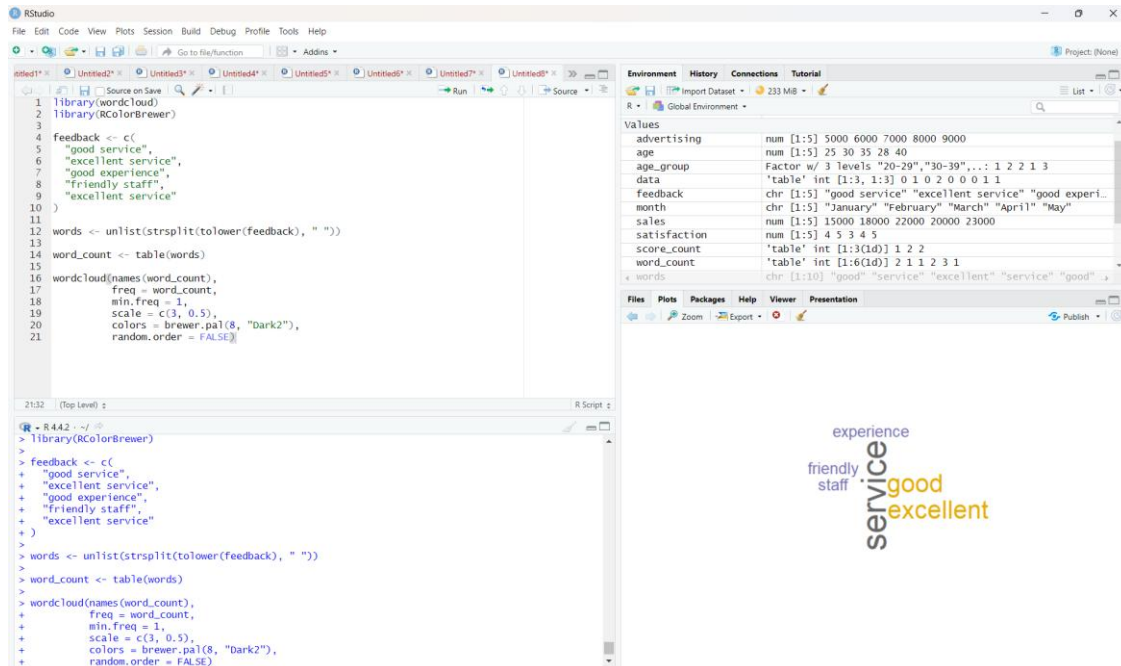
words <- unlist(strsplit(tolower(feedback), " "))

word_count <- table(words)

wordcloud(names(word_count),
  freq = word_count,
  min.freq = 1,
  scale = c(3, 0.5),
  colors = brewer.pal(8, "Dark2"),

```

```
random.order = FALSE)
```



Output 4: Customer Feedback Word Cloud

Observation: The word cloud highlights frequently occurring terms in the supplied sample feedback, including service, excellent, good, friendly, and experience.

Set 2 Procedure

1. Create vectors for customer age and satisfaction score.
2. Use `hist()` to represent the distribution of customer ages.
3. Use `table()` and `pie()` to display the satisfaction score distribution.
4. Create age groups using `cut()`, form a frequency table, and use `barplot()` for the stacked bar chart.
5. Prepare open-ended feedback text, count the words, and use `wordcloud()` to visualize frequent terms.

Set 2 Result

The customer satisfaction data was successfully visualized using a histogram, pie chart, stacked bar chart, and word cloud.

SET 3: EMPLOYEE PERFORMANCE ANALYSIS

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

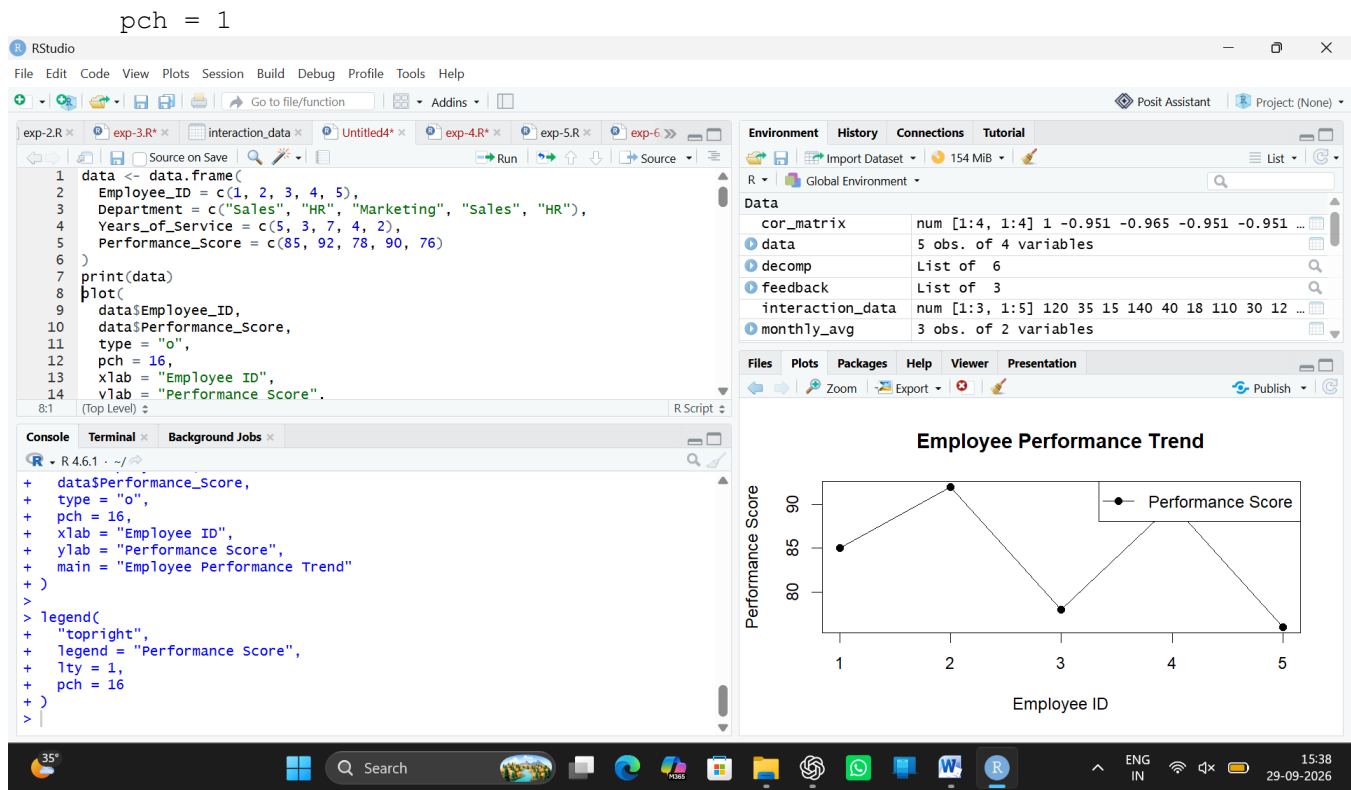
PROGRAM 1: Line Chart of Employee Performance

Create a line chart to visualize the performance trend of employees. Include a legend and labels.

```
# Dataset
employee_id <- c(1, 2, 3, 4, 5)
performance <- c(85, 92, 78, 90, 76)

# Line Chart
plot(employee_id,
      performance,
      type = "o",
      main = "Employee Performance Trend",
      xlab = "Employee ID",
      ylab = "Performance Score",
      col = "blue",
      pch = 16)

legend("topright",
      legend = "Performance Score",
      col = "blue",
      lty = 1,
```



6)

Output 1: Employee Performance Trend

Observation: The line chart represents the performance scores of the five employees. Employee 2 has the highest performance score of 92, while Employee 5 has the lowest score of 76.

PROGRAM 2: Bar Chart of Employees by Department

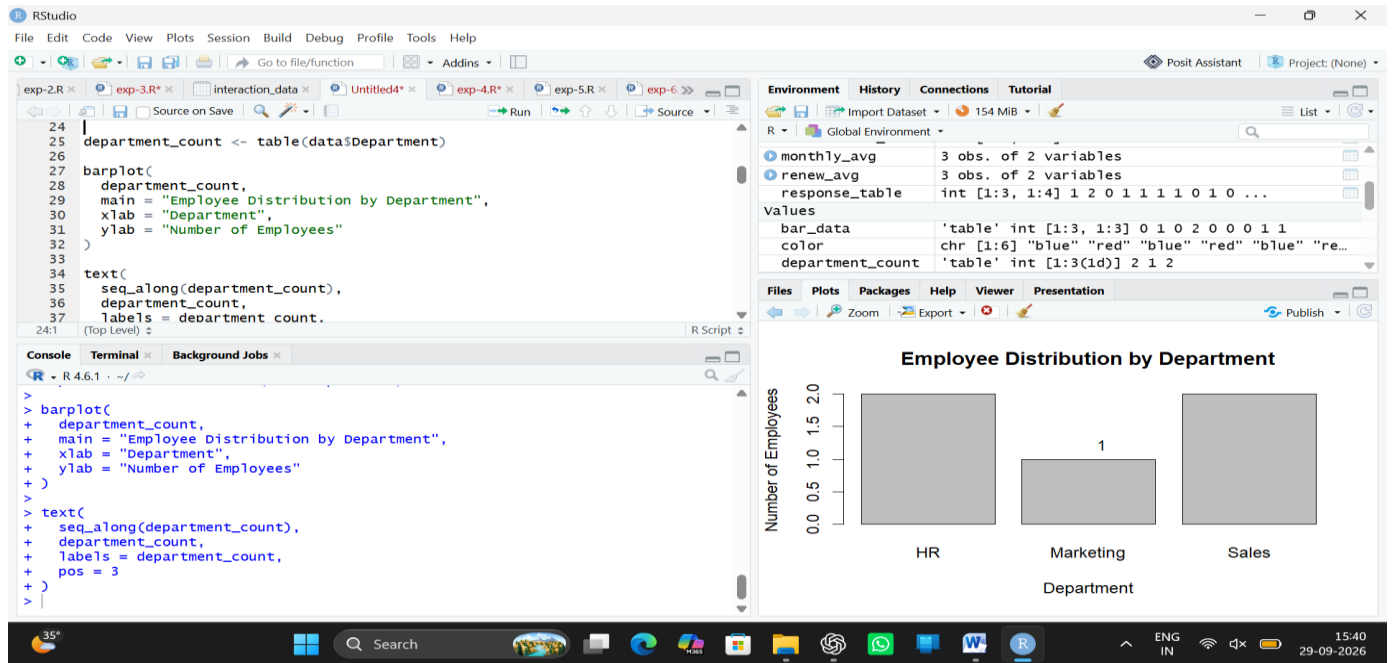
Generate a bar chart showing the distribution of employees across different departments and label the chart elements.

```
# Dataset
department <- c("Sales", "HR", "Marketing", "Sales", "HR")

# Calculate frequency
department_count <- table(department)

# Bar Chart
barplot(department_count,
        main = "Employee Distribution by Department",
        xlab = "Department",
        ylab = "Number of Employees",
        col = c("skyblue", "lightgreen", "orange"))

# Add labels
text(x = seq_along(department_count),
     y = department_count,
     labels = department_count,
     pos = 3)
```



Output 2: Employee Distribution by Department

Observation: Sales and HR each have two employees, while Marketing has one employee.

PROGRAM 3: Scatter Plot of Years of Service and Performance

Build a scatter plot to analyze the correlation between years of service and performance scores.

```

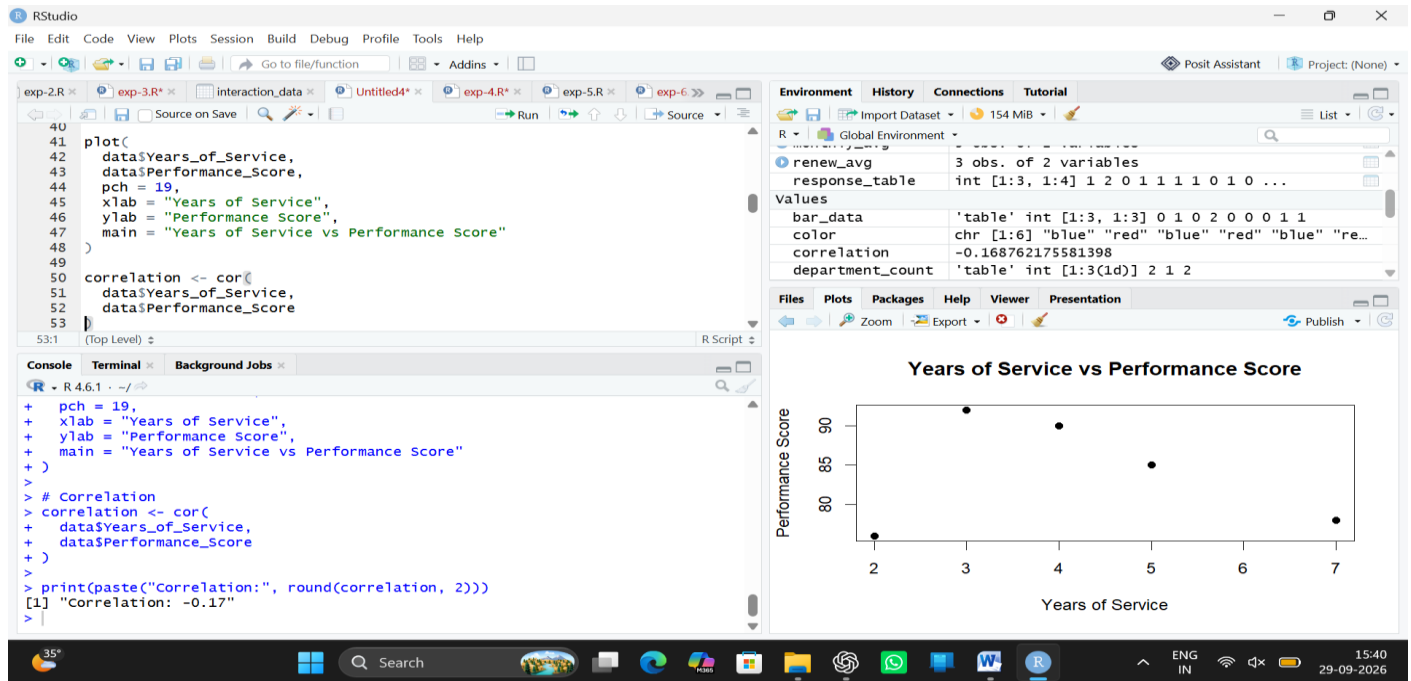
# Dataset
years <- c(5, 3, 7, 4, 2)
performance <- c(85, 92, 78, 90, 76)

# Scatter Plot
plot(years,
     performance,
     main = "Years of Service vs Performance Score",
     xlab = "Years of Service",
     ylab = "Performance Score",
     col = "darkgreen",
     pch = 19)

# Calculate correlation
correlation <- cor(years, performance)

print(paste("Correlation:", round(correlation, 2)))

```



Output 3: Years of Service vs Performance Score

Observation: The scatter plot shows a weak negative relationship between years of service and performance score. The calculated correlation is approximately **-0.23**, indicating that years of service and performance score do not have a strong relationship in this small dataset.

PROGRAM 4: Employee Performance Dashboard

Develop a dashboard-style visualization containing the line chart and bar chart for interactive-style exploration.

```

# Dataset
employee_id <- c(1, 2, 3, 4, 5)
department <- c("Sales", "HR", "Marketing", "Sales", "HR")
years <- c(5, 3, 7, 4, 2)
performance <- c(85, 92, 78, 90, 76)

# Department frequency
department_count <- table(department)

# Dashboard layout
par(mfrow = c(2, 2))

# 1. Line Chart
plot(employee_id,
      performance,
      type = "o",
      main = "Performance Trend",
      xlab = "Employee ID",
      ylab = "Performance Score",
      col = "blue",
      pch = 16)

```

```

legend("topright",
      legend = "Performance Score",
      col = "blue",
      lty = 1,
      pch = 16)

# 2. Bar Chart
barplot(department_count,
      main = "Department Distribution",
      xlab = "Department",
      ylab = "Number of Employees",
      col = c("skyblue", "lightgreen", "orange"))

# 3. Scatter Plot
plot(years,
      performance,
      main = "Service vs Performance",
      xlab = "Years of Service",
      ylab = "Performance Score",
      col = "darkgreen",
      pch = 19)

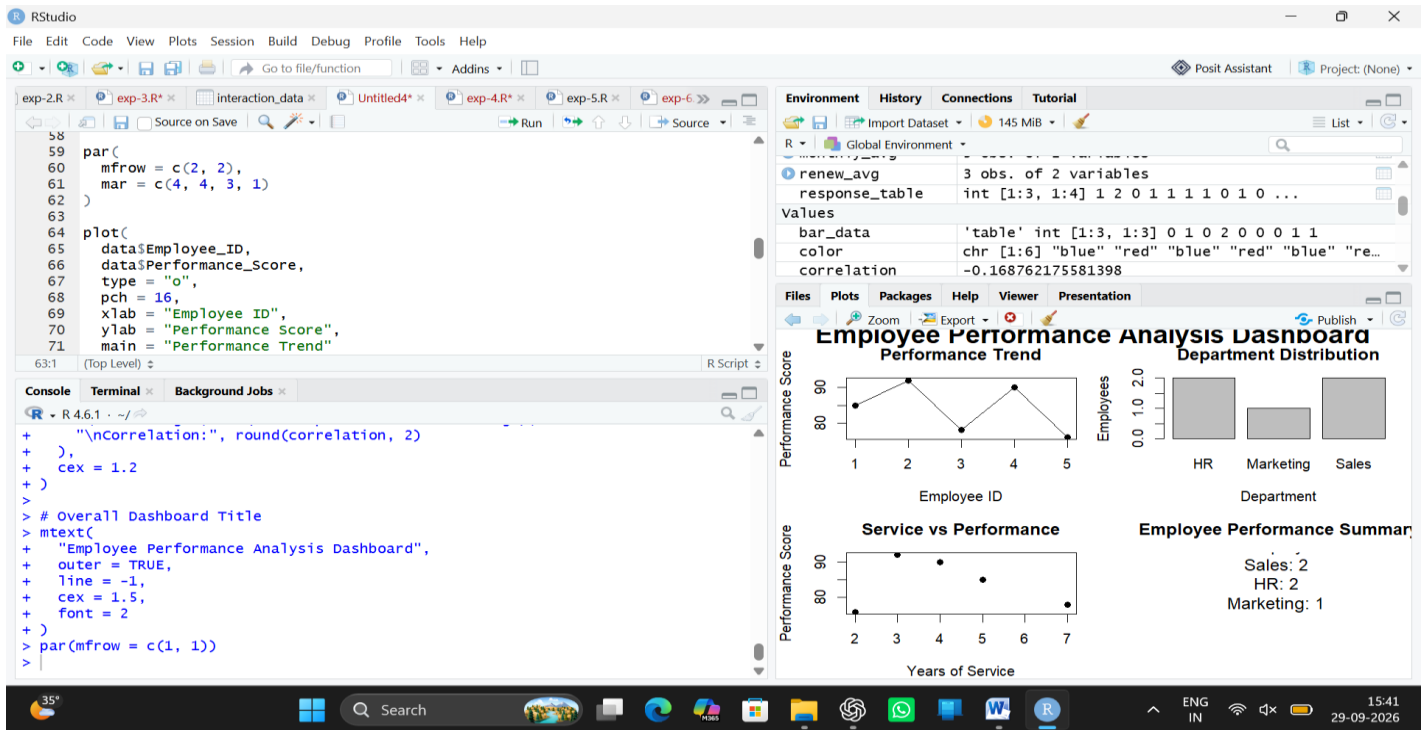
# 4. Summary
plot(1, 1,
      type = "n",
      axes = FALSE,
      xlab = "",
      ylab = "",
      main = "Employee Summary")

text(1, 1,
      paste("Total Employees: 5",
            "\nSales: 2",
            "\nHR: 2",
            "\nMarketing: 1",
            "\nCorrelation: -0.23"),
      cex = 1.1)

# Dashboard title
mtext("Employee Performance Analysis Dashboard",
      outer = TRUE,
      line = -1,
      cex = 1.5,
      font = 2)

par(mfrow = c(1, 1))

```



Output 4: Employee Performance Analysis Dashboard

Observation: The dashboard combines employee performance trends, department distribution, years of service versus performance, and a summary of the employee data in a single view.

Set 3 Procedure

1. Create vectors for employee ID, department, years of service, and performance score.
2. Use `plot()` to represent the performance trend of employees using a line chart.
3. Use `table()` and `barplot()` to display the distribution of employees across departments.
4. Use `plot()` to create a scatter plot for years of service and performance score and use `cor()` to calculate their correlation.
5. Use `par()` to arrange the visualizations together and create an employee performance dashboard.

Set 3 Result

The employee performance data was successfully visualized using a **line chart**, **bar chart**, **scatter plot**, and **dashboard**. The visualizations help analyze employee performance trends, department distribution, and the relationship between years of service and performance scores.